ESCOLA
SUPERIOR
DE TECNOLOGIA
E GESTÃO

P.PORTO

# INTRODUÇÃO ÀS BASES DE DADOS

# História

- Automatização
- Necessidade de processamento de informação
- Necessidade de armazenamento de dados
- Evolução da informatização nas organizações
- Desenvolvimento de aplicações

# Sistemas Baseados em Ficheiros

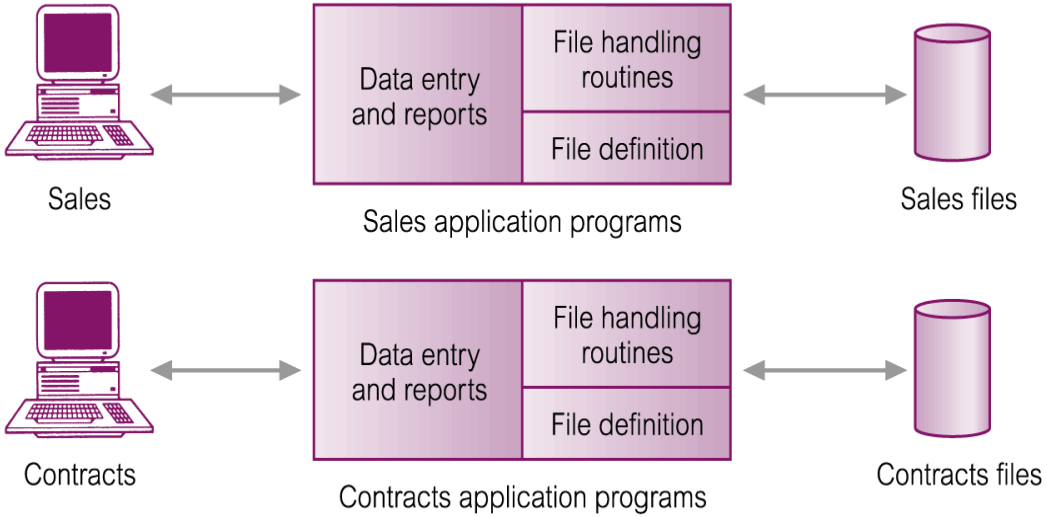



**Figure 1.5**
File-based processing.

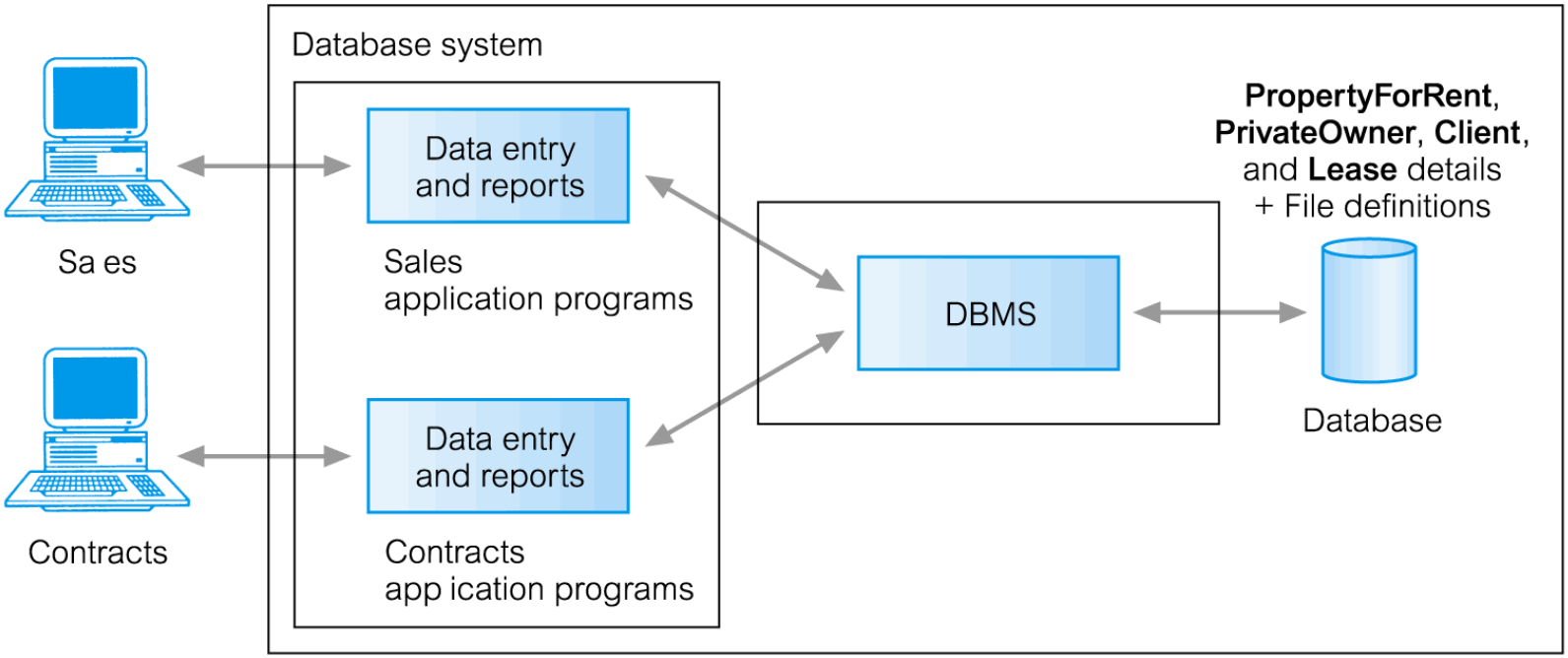
Sales Files

**PropertyForRent** (propertyNo, street, city, postcode, type, rooms, rent, ownerNo)
**PrivateOwner** (ownerNo, fName, lName, address, telNo)
**Client** (clientNo, fName, lName, address, telNo, prefType, maxRent)

Contracts Files

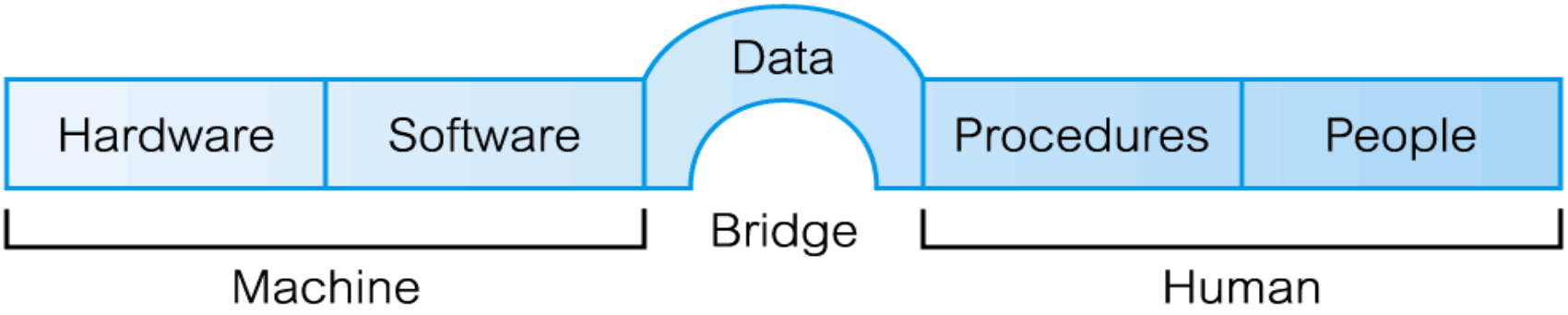
**Lease** (leaseNo, propertyNo, clientNo, rent, paymentMethod, deposit, paid, rentStart, rentFinish, duration)
**PropertyForRent** (propertyNo, street, city, postcode, rent)
**Client** (clientNo, fName, lName, address, telNo)

# Abordagem de Bases de Dados




**PropertyForRent** (propertyNo, street, city, postcode, type, rooms, rent, ownerNo)
**PrivateOwner** (ownerNo, fName, lName, address, telNo)
**Client** (clientNo, fName, lName, address, telNo, prefType, maxRent)
**Lease** (leaseNo, propertyNo, clientNo, paymentMethod, deposit, paid, rentStart, rentF nish)

# Componentes de um Ambiente SGBD



# SGBD

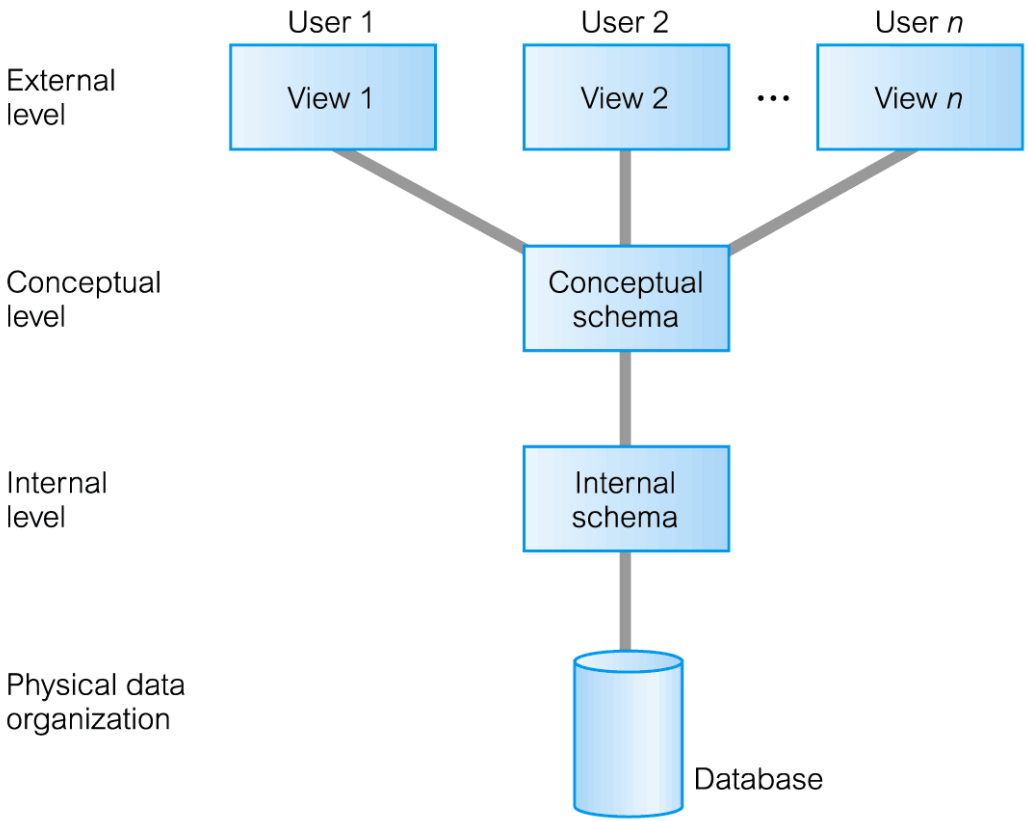
**Vantagens**

- Controlo sobre a redundância de dados
- Consistência de Dados
- Mais informação tendo os mesmo dados
- Partilha de Dados
- Integridade dos Dados melhorada
- Mais Segurança
- Implica uso de standards
- Economia de Escala
- …

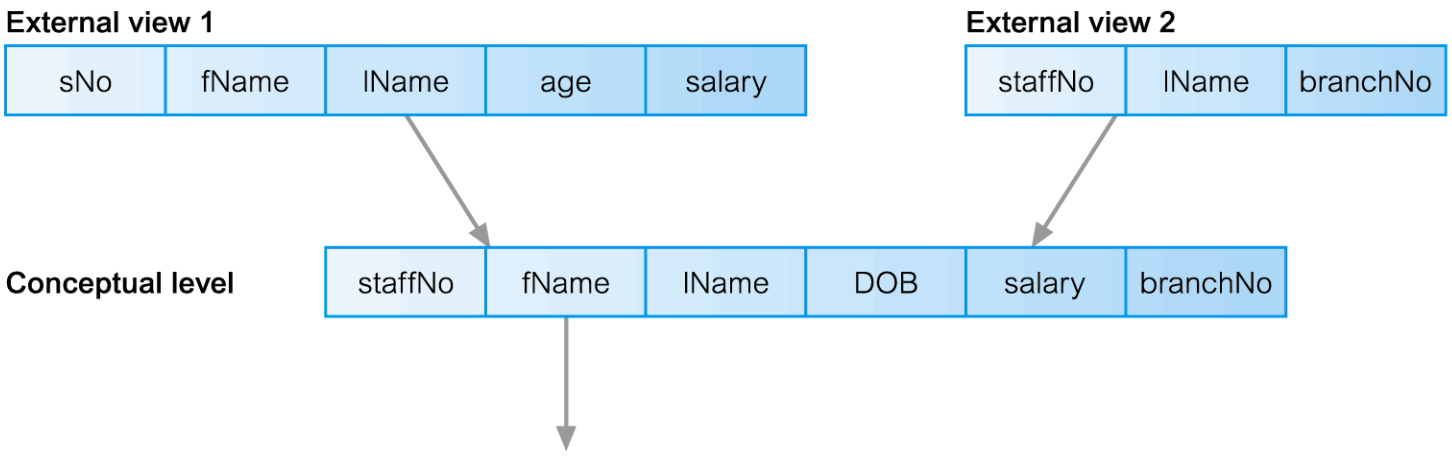
**Desvantagens**

- Complexidade
- Tamanho
- Custo do SGBD
- Custos de hardware acrescidos
- Custo da conversão
- Performance
- Maior impacto em caso de falha
- …

# Arquitetura de três níveis ANSI-SPARC



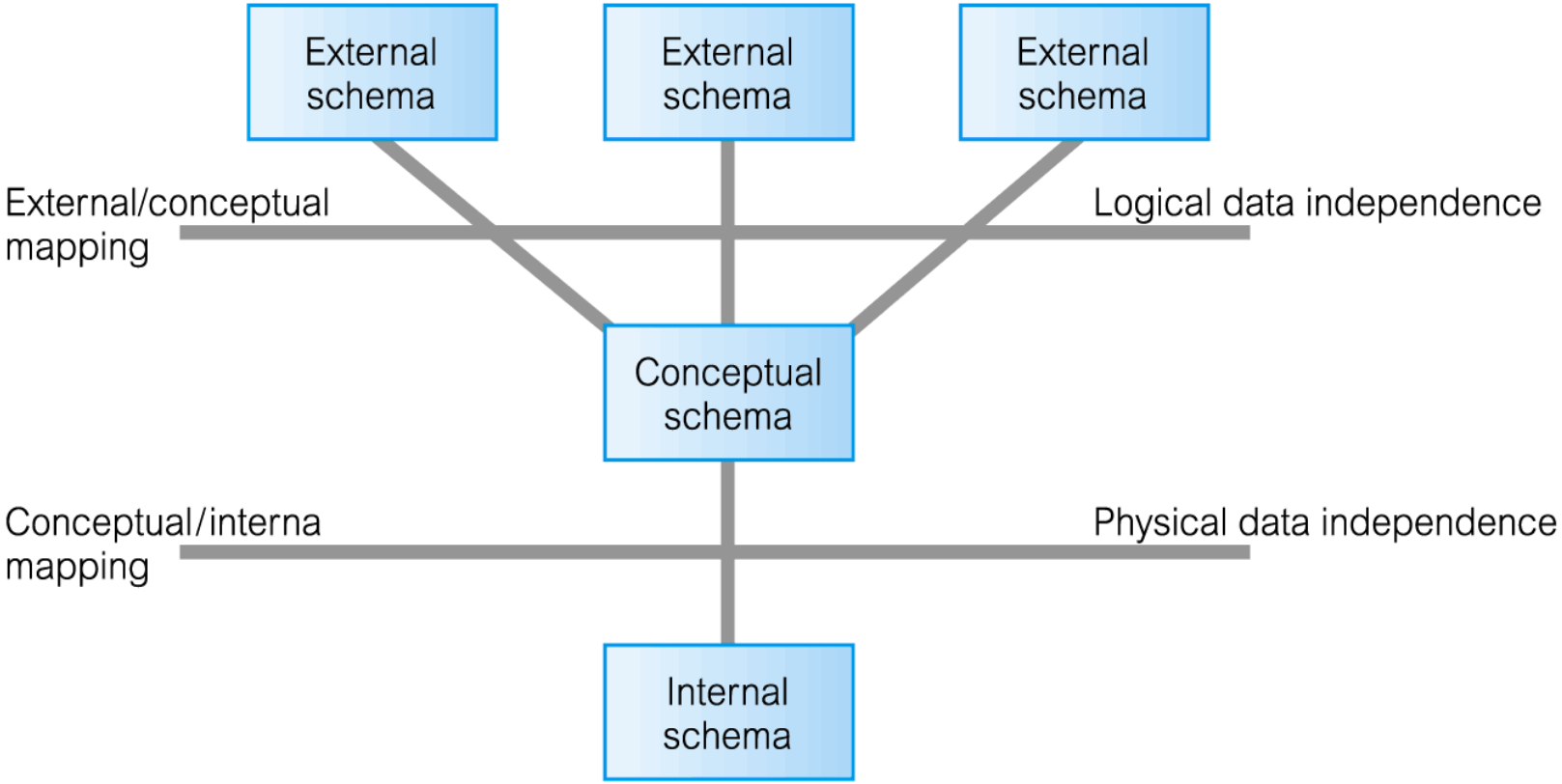
# Diferenças entre os três níveis ANSI-SPARC



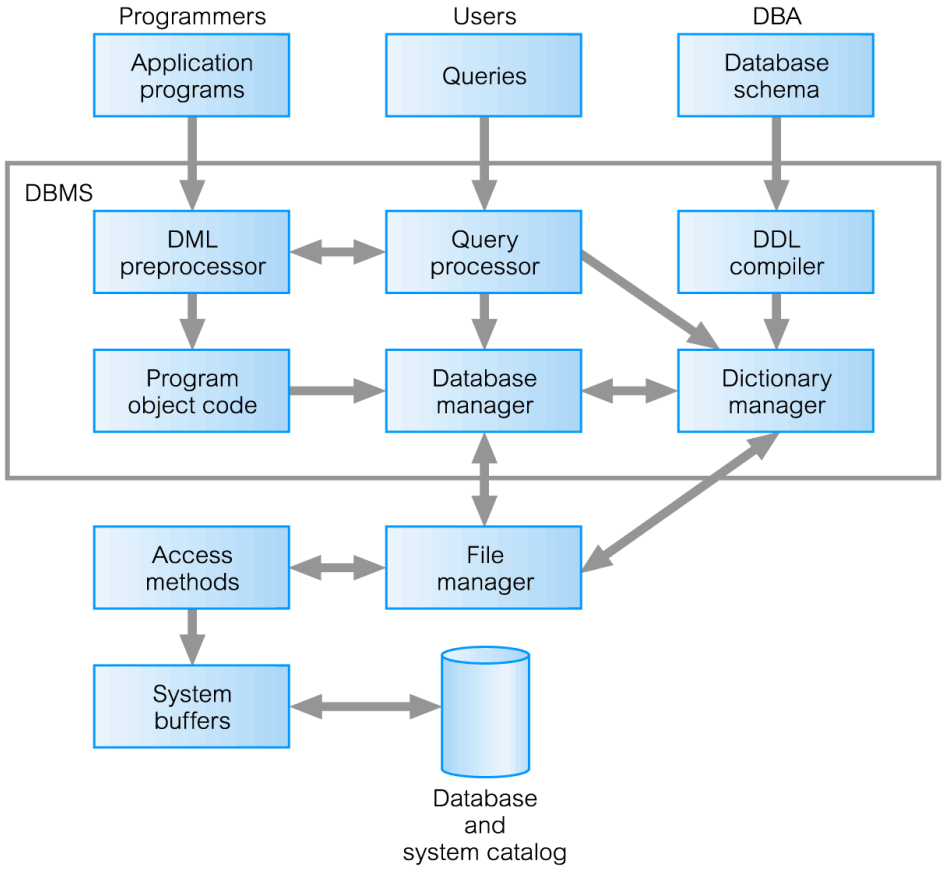

Internal level

```
struct STAFF {
    int staffNo;
    int branchNo;
    char fName [15];
    char lName [15];
    struct date dateOfBirth;
    float salary;
    struct STAFF *next;                /* pointer to next Staff record */
};
index staffNo; index branchNo;         /* define indexes for staff */
```

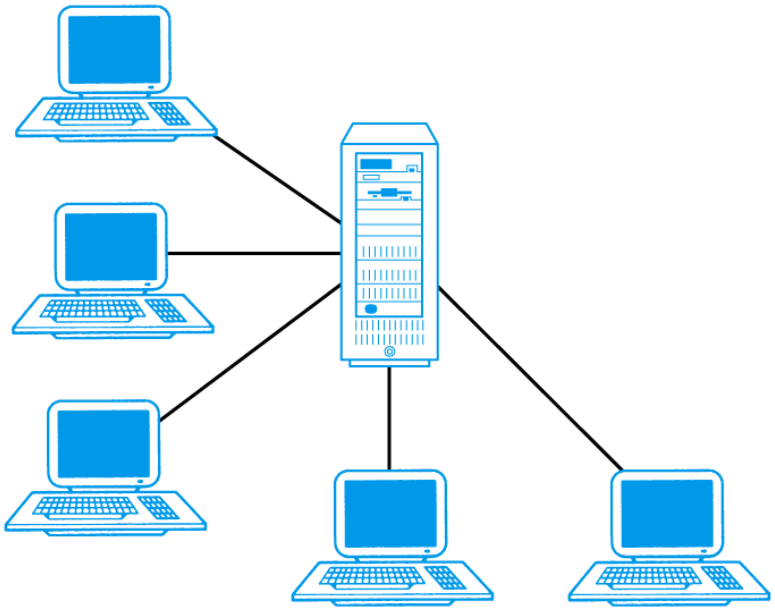
# Independência de dados - ANSI-SPARC



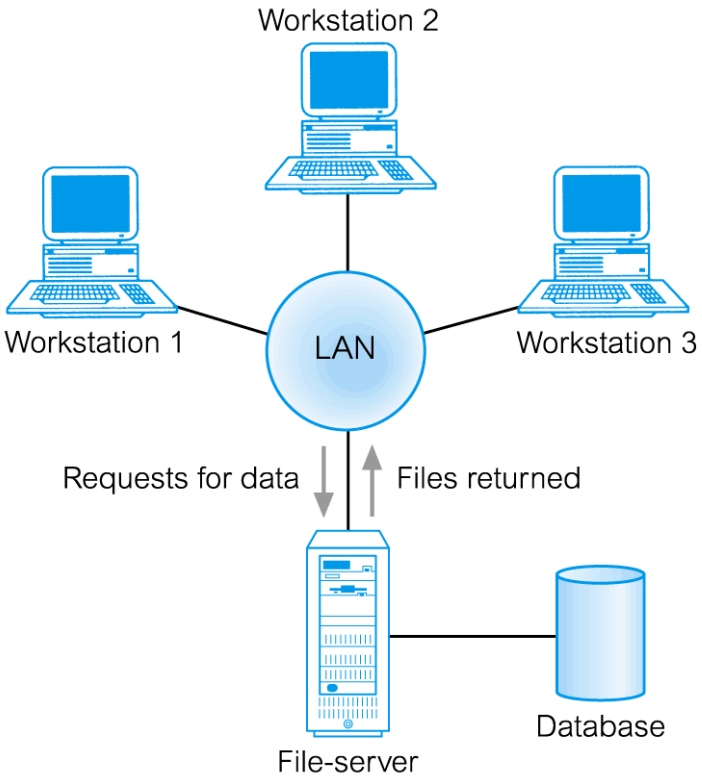
# Componentes de um SGBD



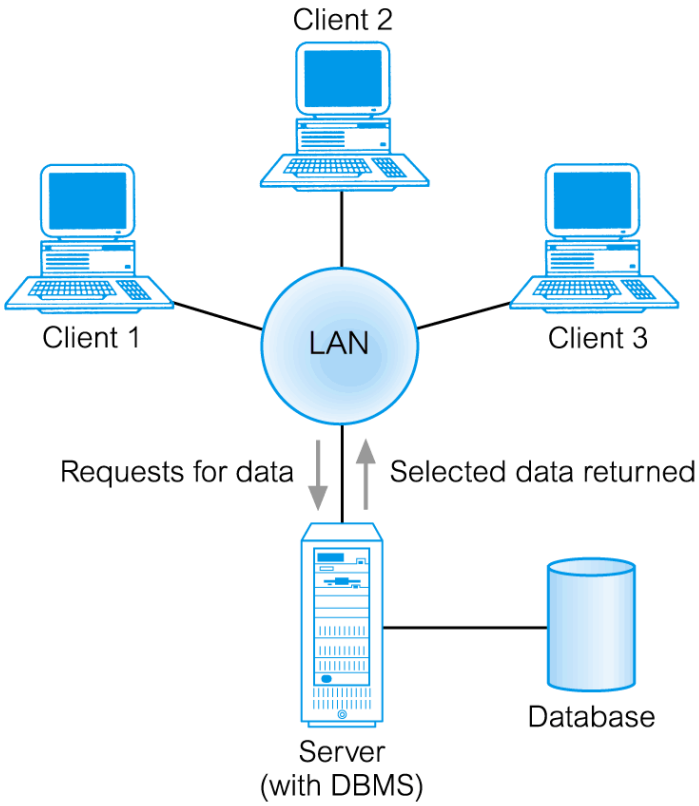
# Arquiteturas SGBD multi-utilizador
# Teleprocessamento



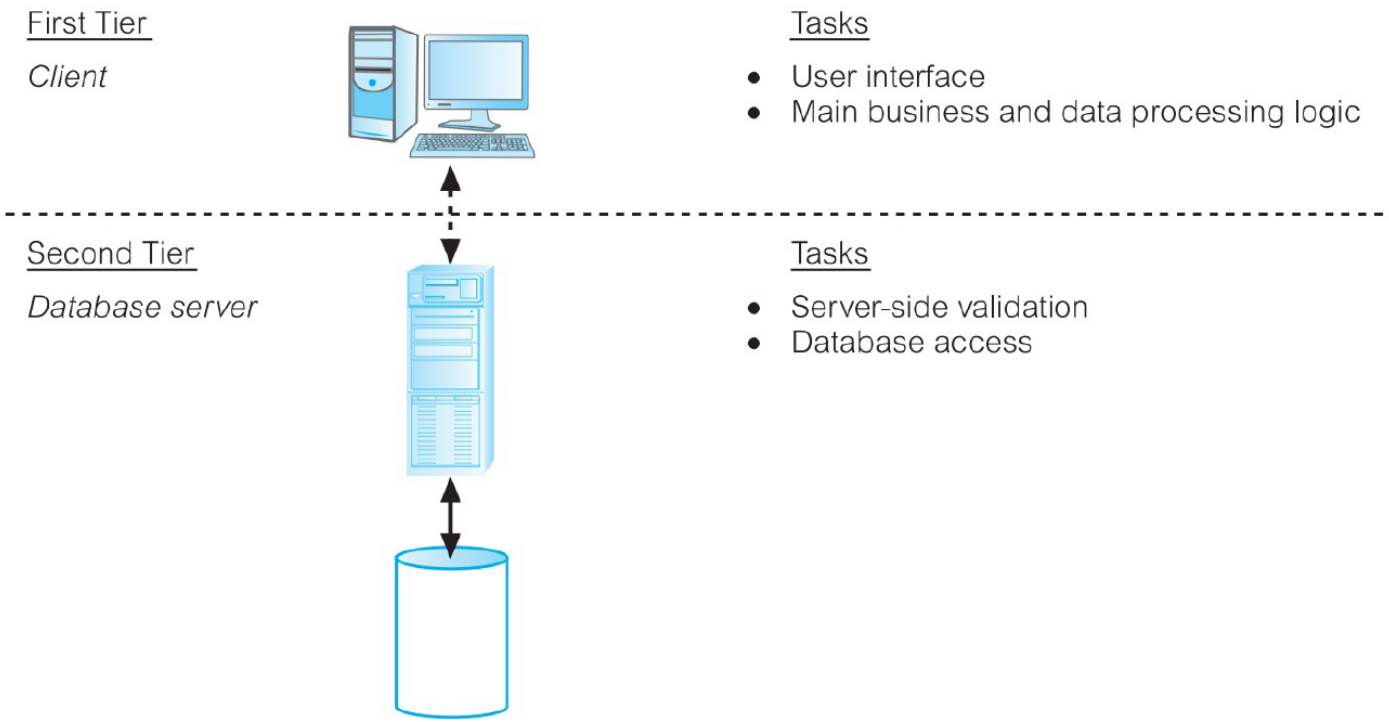
# Arquiteturas SGBD multi-utilizador
# Servidor de Ficheiros



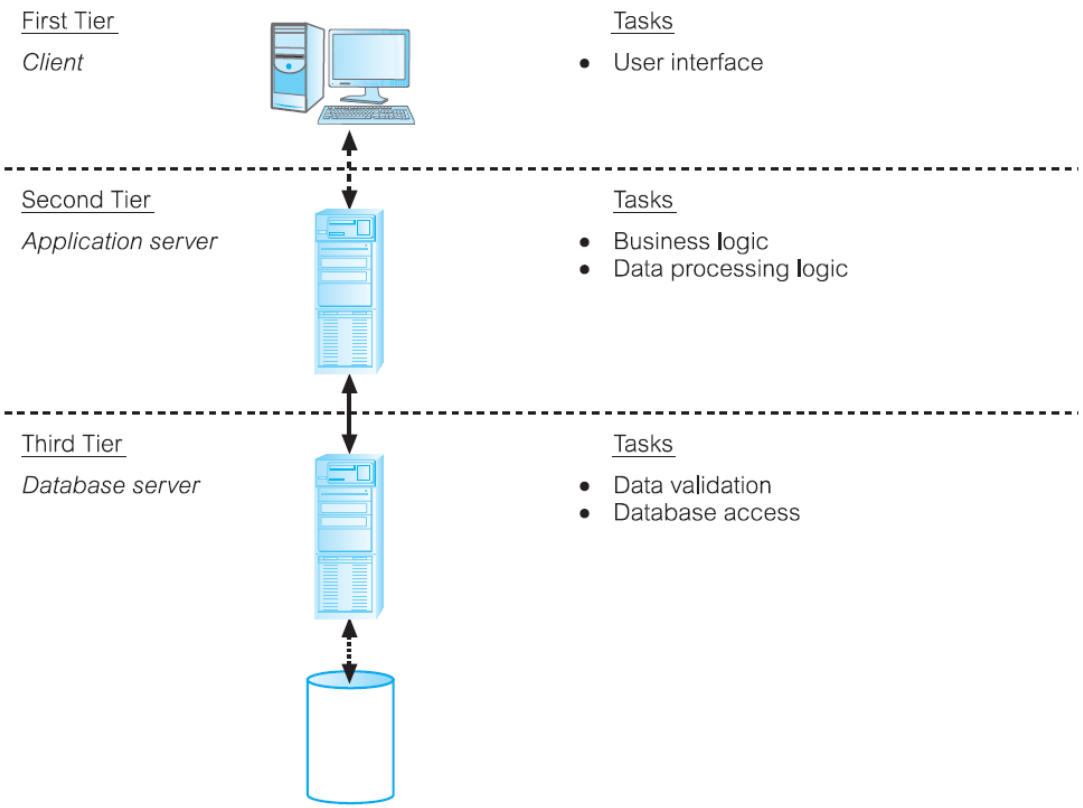
# Arquiteturas SGBD multi-utilizador Cliente /Servidor



# Arquiteturas SGBD multi-utilizador Cliente-Servidor – dois níveis



# Arquiteturas SGBD multi-utilizador Cliente-Servidor – três níveis



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